

Chapter 6

Static Magnetic Fields



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6.1 Fundamental Postulates of Magnetostatics in Free Space

- ❖ The two fundamental postulates of magnetostatics are

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\nabla \cdot \mathbf{B} = 0$$

- where μ_0 is the permeability of free space:

$$\mu_0 = 4\pi \times 10^{-7} \quad (\text{H/m})$$

- And, $\mathbf{J}$ is the current density. Since the divergence of the curl of any vector field is zero, we obtain

$$\nabla \cdot (\nabla \times \mathbf{B}) \equiv 0 \quad \longrightarrow \quad \nabla \cdot \mathbf{J} = 0$$

- which is consistent with the equation of continuity.



• Equations for Steady Magnetic Fields in Free Space (integral form)

❖ Ampere's circuital law

circulation of the magnetic flux density in free space around any closed path is equal to μ_0 times the total current flowing through the surface bounded by the path.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \longrightarrow \quad \oint_l \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

where μ_0 is the permeability of vacuum, and I is the current enclosed by the closed curve.

$$\nabla \cdot \mathbf{B} = 0 \quad \longrightarrow \quad \oint_s \mathbf{B} \cdot d\mathbf{S} = 0$$

1. The total magnetic flux through a **closed** surface is equal to **zero**.
2. The magnetic field lines are **closed everywhere**, with no beginning or end. This may be called the principle of **magnetic flux continuity**.

• Divergence and curl of steady magnetic field

Divergence of constant magnetic field (Differential form)

$$\nabla \cdot \vec{B}(\vec{r}) = 0$$

Curl of constant magnetic field (Differential form)

$$\nabla \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})$$

Principle of continuity of magnetic flux (Integral form)

$$\oint_S \vec{B}(\vec{r}) \cdot d\vec{S} = 0$$

Ampere circuital law (Integral form)

$$\oint_C \vec{B}(\vec{r}) \cdot d\vec{l} = \mu_0 I$$

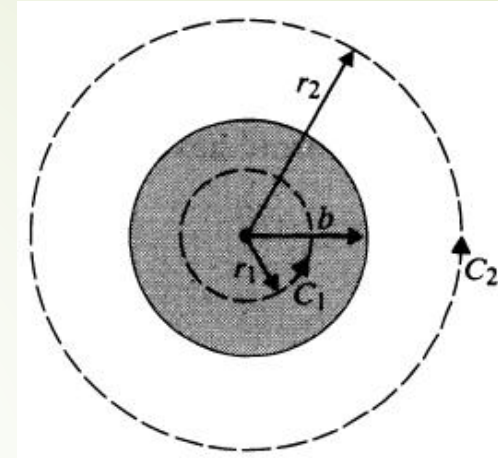
Conclusion:

- A steady magnetic field is a solenoidal and rotational field, which is a non-conservative field
- Current is the vortex source of the magnetic field
- The magnetic flux line is a closed curve with no start and end point

Example

Ex: An infinitely long, straight conductor with a circular cross section of radius b carries a steady current I . Determine the magnetic flux density both inside and outside the conductor.

Solution :



a) Inside the conductor:

$$\mathbf{B}_1 = \mathbf{a}_\phi B_{\phi 1}, \quad d\ell = \mathbf{a}_\phi r_1 d\phi$$
$$\oint_{C_1} \mathbf{B}_1 \cdot d\ell = \int_0^{2\pi} B_{\phi 1} r_1 d\phi = 2\pi r_1 B_{\phi 1}.$$

The current through the area enclosed by C_1 is

$$I_1 = \frac{\pi r_1^2}{\pi b^2} I = \left(\frac{r_1}{b}\right)^2 I.$$

Therefore, from Ampère's circuital law,

$$\mathbf{B}_1 = \mathbf{a}_\phi B_{\phi 1} = \mathbf{a}_\phi \frac{\mu_0 r_1 I}{2\pi b^2}, \quad r_1 \leq b. \quad (6-11a)$$



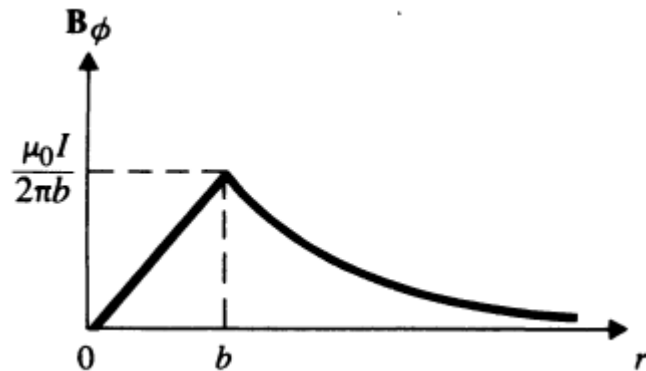
b) Outside the conductor:

$$\mathbf{B}_2 = \mathbf{a}_\phi B_{\phi 2}, \quad d\ell = \mathbf{a}_\phi r_2 d\phi$$
$$\oint_{C_2} \mathbf{B}_2 \cdot d\ell = 2\pi r_2 B_{\phi 2}.$$

Path C_2 outside the conductor encloses the total current I . Hence

$$\mathbf{B}_2 = \mathbf{a}_\phi B_{\phi 2} = \mathbf{a}_\phi \frac{\mu_0 I}{2\pi r_2}, \quad r_2 \geq b. \quad (6-11b)$$

Examination of Eqs. (6-11a) and (6-11b) reveals that the magnitude of $\mathbf{B}$ increases linearly with r_1 from 0 until $r_1 = b$, after which it decreases inversely with r_2 . The variation of B_ϕ versus r is sketched in Fig. 6-2(b).



Magnetic flux density of an infinitely long circular conductor carrying a current I

If it is not a solid cylindrical conductor but a very thin circular tube carrying a surface current density J_s , then it is obvious:

$$B = \begin{cases} 0, & r < b, \\ \mathbf{a}_\phi \frac{\mu_0 b}{r} J_s, & r > b. \end{cases}$$



6.2 Vector Magnetic Potential

- ➡ The divergence-free postulate of $\mathbf{B}$, assures that $\mathbf{B}$ is solenoidal.

$$\nabla \cdot \mathbf{B} = 0$$

- ❖ As a consequence, $\mathbf{B}$ can be expressed as the curl of another vector field, say $\mathbf{A}$, such that

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (T) \quad \longleftarrow \quad \nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$$

A: vector magnetic potential

- ❖ The definition of a vector field requires the specification of both its curl and its divergence.
- ❖ How do we choose divergence of $\mathbf{A}$?

The divergence of the magnetic vector potential can be chosen arbitrarily!

1. Definition of vector magnetic potential

$$\nabla \cdot \vec{A} = 0 \quad (\text{steady magnetic field})$$



Coulomb gauge

2. Differential equations of magnetic vector potential (general problems)

$$\left. \begin{aligned} \vec{B} &= \nabla \times \vec{A} \\ \nabla \times \vec{B} &= \mu_0 \vec{J} \end{aligned} \right\} \Rightarrow \nabla \times \nabla \times \vec{A} = \mu_0 \vec{J} \Rightarrow \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\nabla \cdot \vec{A} = 0$$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J} \quad \text{--- Vector Poisson equation}$$

In the source free area :

$$\vec{J} = 0 \quad \Rightarrow \quad \nabla^2 \vec{A} = 0 \quad \text{--- Vector Laplace's Equation}$$

3. The problems in infinite uniform medium space (special problems)

➤ The analogy method to solve

$$\begin{aligned}\vec{A} &= A_1 \vec{e}_x + A_2 \vec{e}_y + A_3 \vec{e}_z \\ \vec{J} &= J_1 \vec{e}_x + J_2 \vec{e}_y + J_3 \vec{e}_z\end{aligned}$$

$$\nabla^2 \varphi(\vec{r}) = -\frac{1}{\varepsilon} \rho(\vec{r})$$

$$\varphi(\vec{r}) = \frac{1}{4\pi\varepsilon} \int_{V'} \frac{\rho(\vec{r}') dv'}{R} + C$$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}$$



$$\begin{cases} \nabla^2 A_1 = -\mu_0 J_1 \\ \nabla^2 A_2 = -\mu_0 J_2 \\ \nabla^2 A_3 = -\mu_0 J_3 \end{cases}$$

$$A_i(\vec{r}) = \frac{\mu_0}{4\pi} \int_{V'} \frac{J_i(\vec{r}') dv'}{R} + C_i$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}(\vec{r}') dv'}{R} + \vec{C}$$



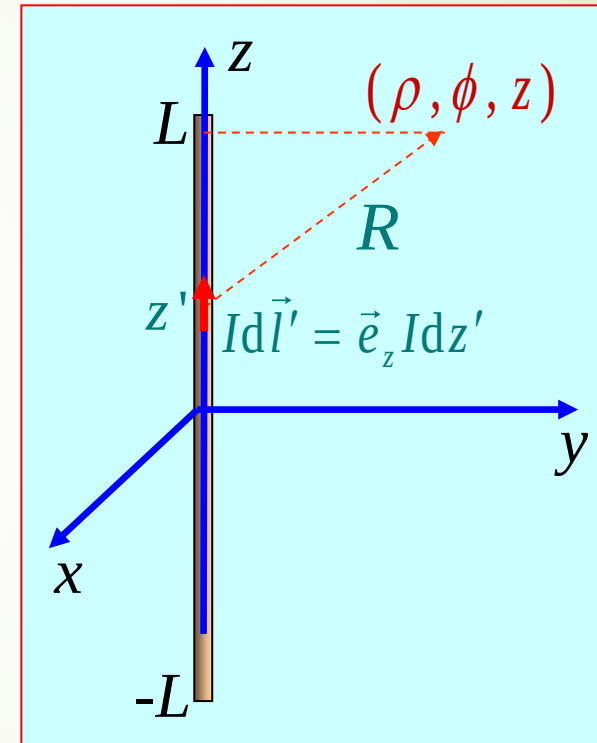
Ex. Find the magnetic vector potential of an infinite long line carrying current I long z axis.

Solution: the magnetic vector potential for $2L$ line, current element $Id\vec{l}' = \vec{e}_z Idz'$

$$\begin{aligned} \text{to } P(\rho, \phi, z), \text{ the } R &= \sqrt{\rho^2 + (z - z')^2} \\ \vec{A}(\vec{r}') &= \vec{e}_z \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{1}{\sqrt{\rho^2 + (z - z')^2}} dz' \\ &= \vec{e}_z \frac{\mu_0 I}{4\pi} \ln[z' - z + \sqrt{\rho^2 + (z - z')^2}] \Big|_{-L}^L \\ &= \vec{e}_z \frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{\rho^2 + (z - L)^2} - (z - L)}{\sqrt{\rho^2 + (z + L)^2} - (z + L)} \end{aligned}$$

The magnetic vector potential for $L \rightarrow \infty$

$$\vec{A}(\vec{r}) = \vec{e}_z \frac{\mu_0 I}{2\pi} \ln \frac{2L}{\rho} + C$$



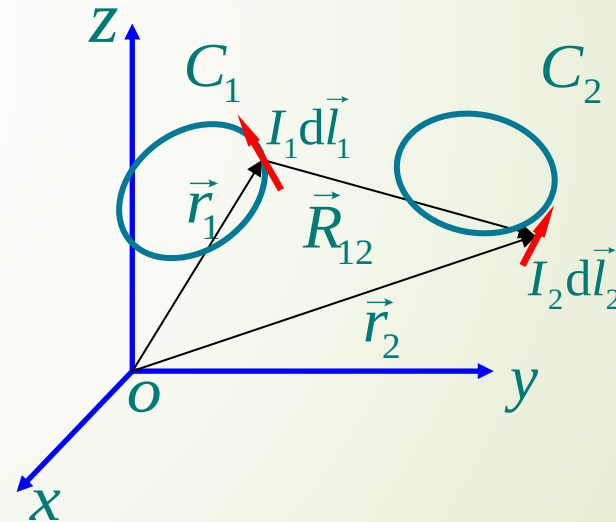
6.3 The Biot-Savart Law and Applications

In static magnetic field: The field has the effect on the current element located in a magnetic field.

1. Ampere's force law

$$\vec{F}_{12} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{I_2 d\vec{l}_2 \times (I_1 d\vec{l}_1 \times \vec{R}_{12})}{R_{12}^3}$$

Practice: $\vec{F}_{21} = -\vec{F}_{12}$



Discussion

- Ampère's force law obeys Newton's third law (the law of action and reaction).
- Ampère's Law describes the existence of forces between current-carrying circuits, but it does not specify how these forces are exerted.

2. Magnetic flux density $\vec{B}$

According to Ampere's force law

**Biot-Savart
Law**

$$\vec{F}_{12} = \oint_{C_2} I_2 d\vec{l}_2 \times \left(\frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{R}_{12}}{R_{12}^3} \right) = \oint_{C_2} I_2 d\vec{l}_2 \times \vec{B}_1(\vec{r}_2)$$

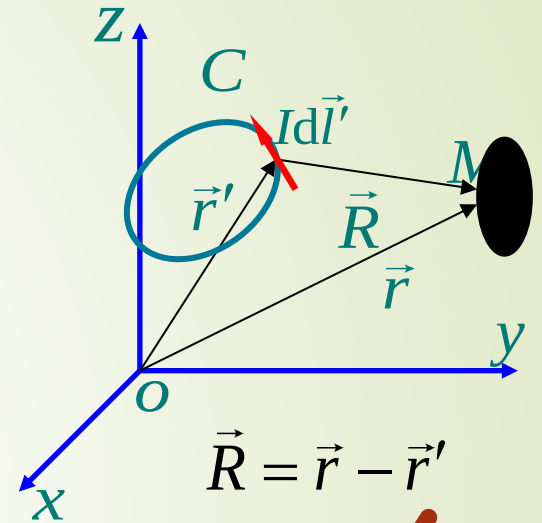
which: $\vec{B}_1(\vec{r}_2) = \frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{R}_{12}}{R_{12}^3}$ is independent of I_2 , we call it magnetic flux density generated by I_1 , and its unit is T.

- *Biot-Savart Law describes how the magnetic force are exerted*
——by magnetic field

Magnetic flux density of different current models

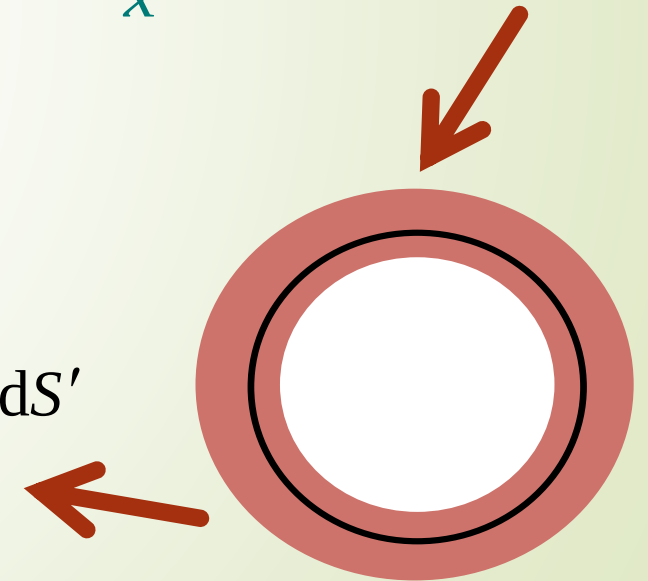
🏠 Magnetic flux density produced by arbitrary current loop C

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{Id\vec{l}' \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_C \frac{Id\vec{l}' \times \vec{R}}{R^3}$$



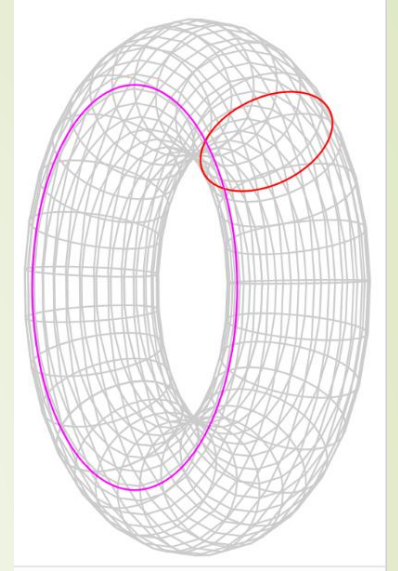
🏠 Magnetic flux density produced by surface current

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{J}_s ds \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_S \frac{J_s(\vec{r}') \times \vec{R}}{R^3} dS'$$



🏠 **Magnetic flux density produced by volume current**

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{J} dV \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{J(\vec{r}') \times \vec{R}}{R^3} dV'$$



🏠 **Magnetic flux density produced by convection current**

$$d\vec{B}(\vec{r}) = \frac{\mu_0 \rho \vec{v} \times \vec{R}}{4\pi R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\rho \vec{v} \times \vec{R}}{R^3} dV'$$

3. Magnetic flux densities of several typical current distributions

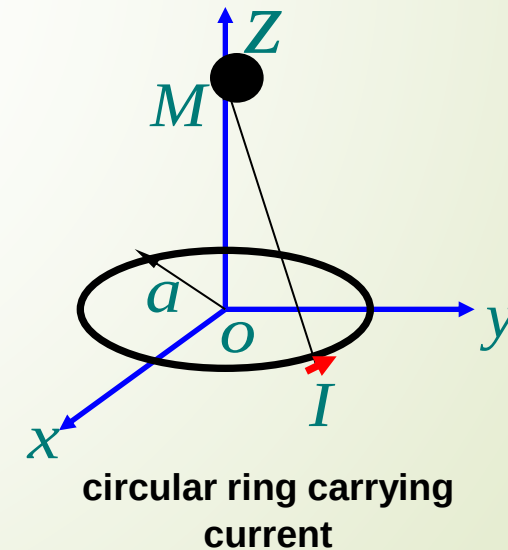
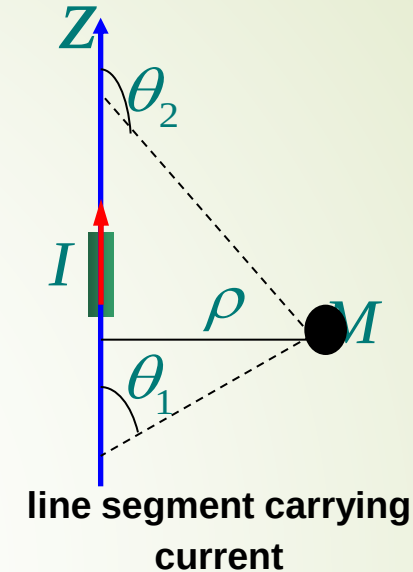
- The magnetic flux density of current carrying line segment :

$$\vec{B} = \vec{e}_\phi \frac{\mu_0 I}{4\pi\rho} (\cos \theta_1 - \cos \theta_2) \quad (\text{Finite long})$$

$$\vec{B} = \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho} \quad (\text{Infinite long})$$

- Magnetic flux density on the axis of current carrying circular ring :

$$\vec{B}(0, 0, z) = \vec{e}_z \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}$$



Ex. Calculate the magnetic flux density at arbitrary point at the center of the circular ring.

Solution : Establish a best coordinate system ,as shown in Figure .

For the magnetic flux density at arbitrary point on the axis $P (0, 0, z)$, because:

$$Id\vec{l}' = \vec{e}_\phi I a d\phi' \quad \vec{r}' = \vec{e}_\rho a \quad \vec{r} = \vec{e}_z z$$

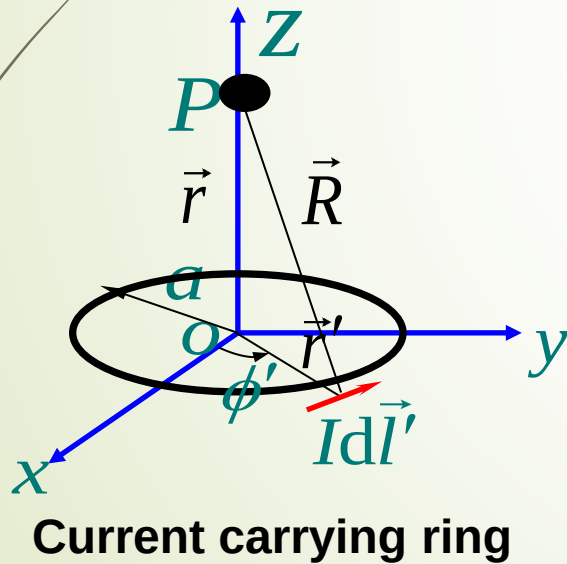
so

$$\vec{r} - \vec{r}' = \vec{e}_z z - \vec{e}_\rho a, \quad |\vec{r} - \vec{r}'| = (z^2 + a^2)^{1/2}$$

$$\begin{aligned} Id\vec{l}' \times (\vec{r} - \vec{r}') &= \vec{e}_\phi I a d\phi' \times (\vec{e}_z z - \vec{e}_\rho a) \\ &= \vec{e}_\rho I a z d\phi' + \vec{e}_z I a^2 d\phi' \end{aligned}$$

So: Magnetic flux density at $P (0, 0, z)$ is

$$\vec{B}(z) = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{\vec{e}_\rho z + \vec{e}_z a}{(z^2 + a^2)^{3/2}} d\phi'$$



for $\int_0^{2\pi} \vec{e}_\rho d\phi' = \int_0^{2\pi} (\vec{e}_x \cos\phi' + \vec{e}_y \sin\phi') d\phi' = 0$ **so**

$$\vec{B}(z) = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{\vec{e}_z a}{(z^2 + a^2)^{3/2}} d\phi' = \vec{e}_z \frac{\mu_0 I a^2}{2(z^2 + a^2)^{3/2}}$$

Discussion

- **Why the magnetic field at P only has Z component ?**
- **Where is the maximum magnetic flux density ?**

At the center of the ring , as $z = 0$: $\vec{B}(0) = \vec{e}_z \frac{\mu_0 I}{2a}$

- **What about the far field? (When $z \gg a$)**

$$(z^2 + a^2)^{3/2} \approx z^3 \qquad \vec{B} = \vec{e}_z \frac{\mu_0 I a^2}{2z^3}$$

Laws and methods of steady magnetic field to the methods of analyzing problems

Laws

- Current sources

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{R}}{R^3}$$

- Vortex source of the magnetic field

$$\nabla \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})$$

- The circulation of the magnetic field is proportional to the current

$$\oint_C \vec{B}(\vec{r}) \cdot d\vec{l} = \mu_0 I$$

Methods

- Known current distribution, solve the distribution of magnetic field

Superposition principle

- Known the distribution of magnetic field

Conduct differential operation

- Known current distribution

Solving differential equations

- Known magnetic field distribution, solve its circulation

Solving integral equations

- Known current, solve its magnetic field

Solving integral equations

- A simple way to solve magnetic flux density
--using **Ampere's circuital law**

Condition: Problem has symmetry, the integral equation can be transformed to algebraic equation !

Example: When surface current density is $\vec{J}_S = \vec{e}_z J_{S0}$, find magnetic flux intensity produced by infinite current sheet.

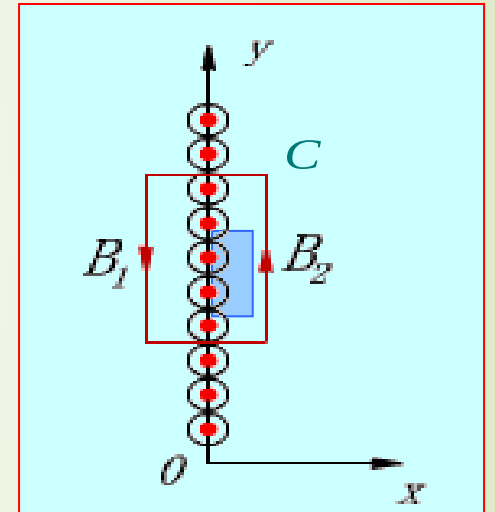
Solution: Establish a best coordinate system

so : $\vec{B}(\vec{r}) = \vec{B}(x)$

Based on the symmetry, the loop integral is:

$$\oint_C \vec{B} \cdot d\vec{l} = B_1 l + B_2 l = \mu_0 J_{S0} l \quad \Rightarrow \quad \vec{B} = \begin{cases} \vec{e}_y \frac{\mu_0 J_{S0}}{2} & x > 0 \\ -\vec{e}_y \frac{\mu_0 J_{S0}}{2} & x < 0 \end{cases}$$

$$B_1 = B_2 = B$$



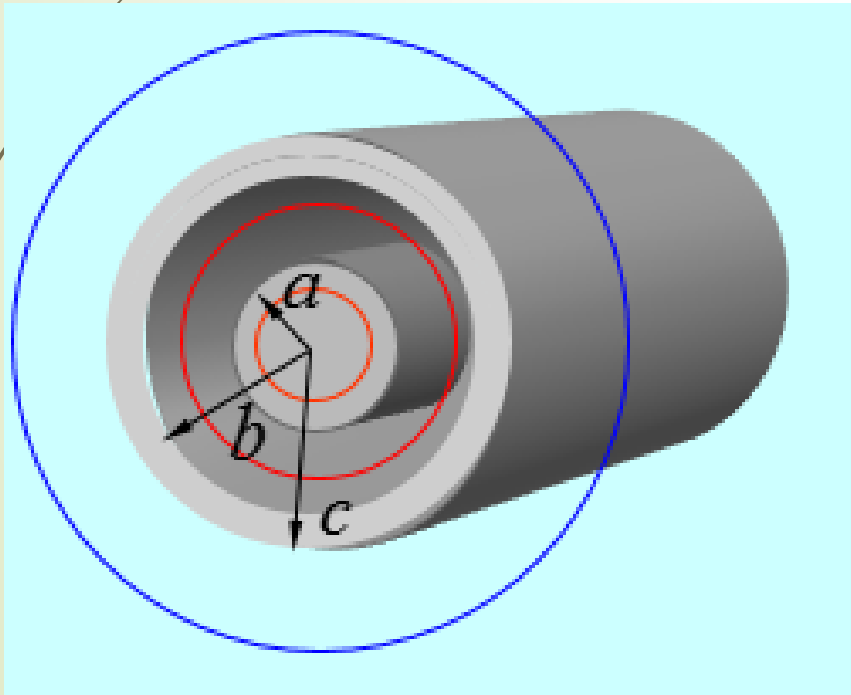
Ex. Find the magnetic flux density of the infinite long current carrying coaxial cable.

Solution: Choose cylindrical coordinate system, so $\vec{B}(\vec{r}) = \vec{e}_\phi B(\rho)$

(1) $0 \leq \rho < a$

Choose ampere loop ($\rho < a$), the current of the cross link is

$$I_1 = \frac{I}{\pi a^2} \cdot \pi \rho^2 = I \frac{\rho^2}{a^2}$$



Applying **Ampere's circuital law**

$$2\pi\rho B_1 = \mu_0 \frac{I\rho^2}{a^2}$$



$$\vec{B}_1 = \vec{e}_\phi \frac{\mu_0 I \rho}{2\pi a^2}$$

(2) $a \leq \rho < b$

$$2\pi\rho B_2 = \mu_0 I \quad \Rightarrow \quad \vec{B}_2 = \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho}$$

(3) $b \leq \rho < c$

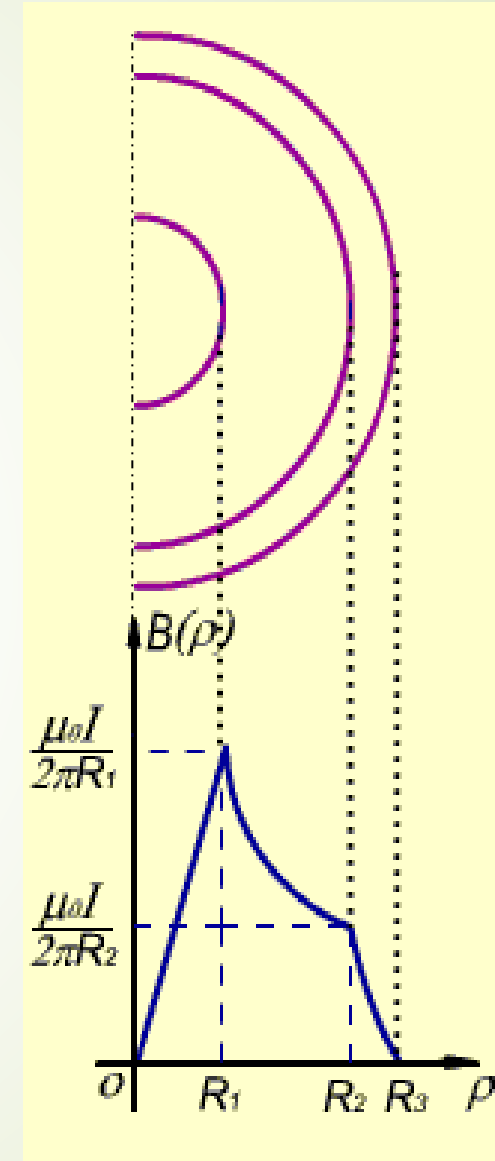
$$I_3 = I - I \frac{\rho^2 - b^2}{c^2 - b^2} = I \frac{c^2 - \rho^2}{c^2 - b^2}$$

Applying **Ampere's circuital law**

$$2\pi\rho B_3 = \frac{\mu_0 I (c^2 - \rho^2)}{c^2 - b^2} \quad \Rightarrow \quad \vec{B}_3 = \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho} \cdot \frac{c^2 - \rho^2}{c^2 - b^2}$$

(4) $c \leq \rho < \infty$

$$I_4 = 0 \quad \Rightarrow \quad \vec{B}_4 = 0$$



6.4 Magnetization and Equivalent Current Densities

1. Magnetization of medium

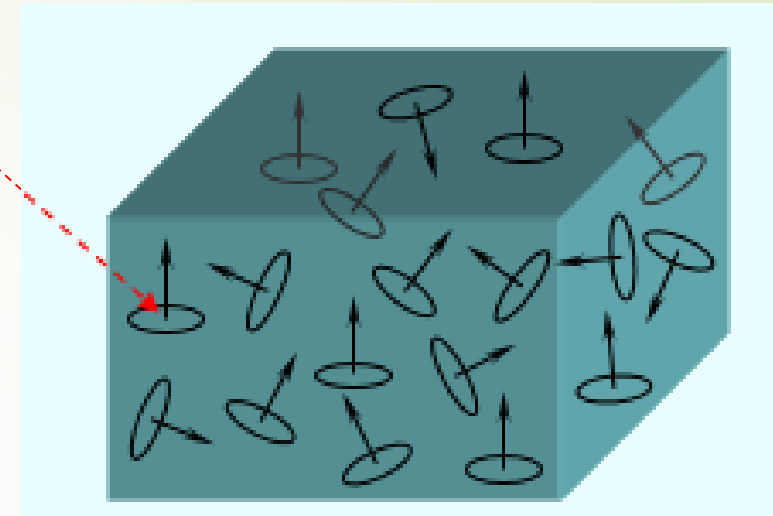
Conclusion

- 1) Under the effect of external magnetic field, the magnetic moments of the medium molecules are oriented, resulting in the magnetic phenomenon (showing magnetism).
- 2) The magnitude of the magnetization is determined by the number of magnetic moment in the medium.

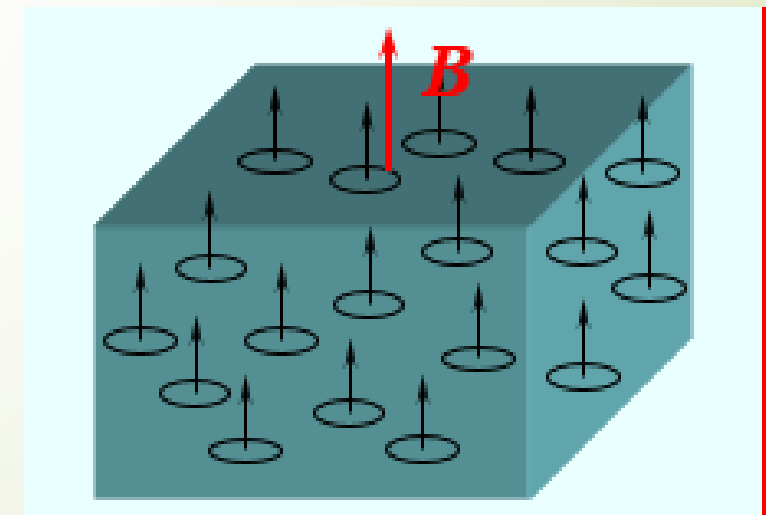
Definition of magnetic moment

$$\vec{p}_e = q_e \vec{l} \quad \vec{p}_m = i \Delta \vec{S}$$

$$\vec{p}_m = i \Delta \vec{S}$$



No external magnetic field

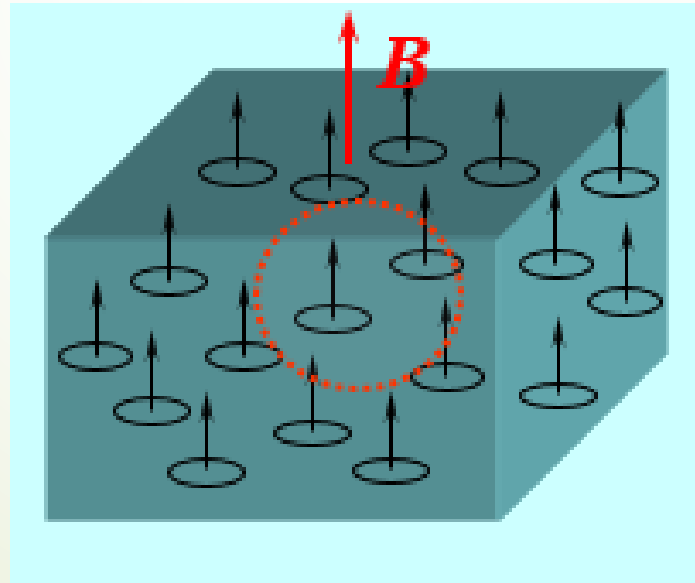


External magnetic field applied

2. Magnetization vector

Definition: Sum of the magnetic dipole moment per unit volume of the medium, that

$$\vec{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_i \vec{p}_{mi}}{\Delta v} \quad (A/m)$$

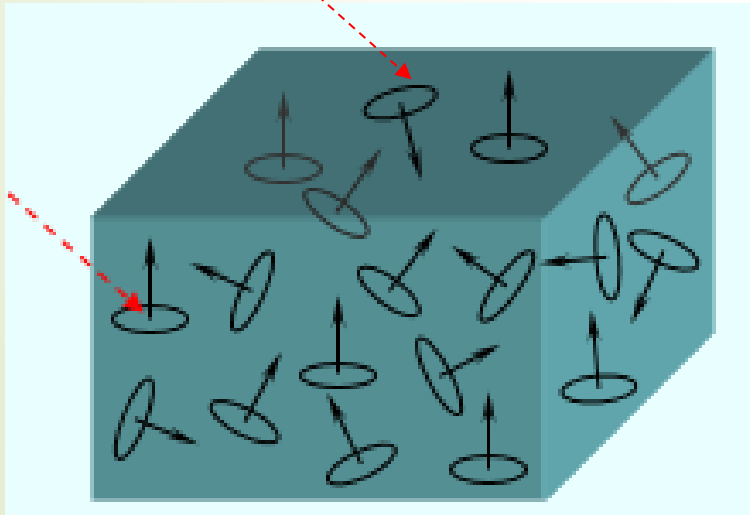


Discussion

1) When the medium has no external magnetic field

$$\sum_i \vec{p}_{mi} = 0 \Rightarrow \vec{M} = 0$$

$$\vec{p}_m = i\Delta\vec{S}$$



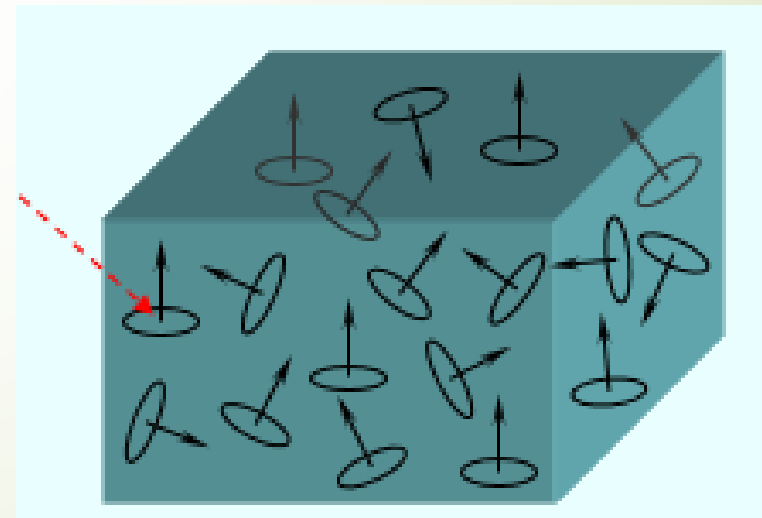
No external magnetic field

2) With external magnetic field

$$\vec{p}_{m1} = \vec{p}_{m2} = \dots = \vec{p}_m$$
$$\Rightarrow \sum_i \vec{p}_{mi} = N\vec{p}_m$$
$$\Rightarrow \vec{M} = n\vec{p}_m$$

There, n is the number of magnetized molecules in the unit volume

$$\vec{M} = n\vec{p}_m$$



No external magnetic field

Conclusion

1) The magnitude of the magnetization vector is related to the dielectric material

2) The magnitude of the magnetization is also related to the intensity of the applied field $\vec{B}_e$

- After the medium is magnetized, extra magnetic field $\vec{B}_m$ will be generated in the space

- The total magnetic field $\vec{B}$ in the medium inside and outside space is

$$\vec{B} = \vec{B}_e + \vec{B}_m$$

Experimental results:

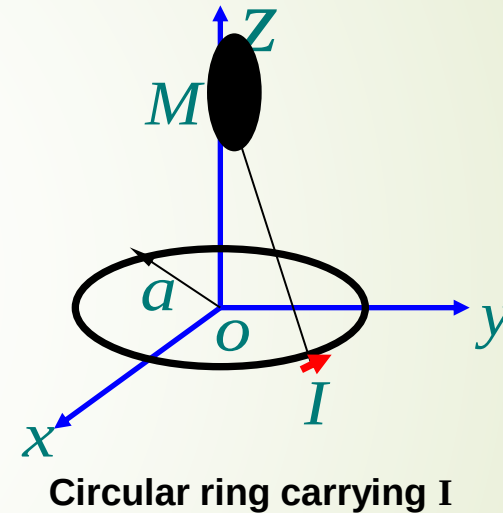
For linear and isotropic media , $\vec{M}$ is proportional to $\vec{H}$, that $\vec{M} = \chi_m \vec{H}$

where, $\chi_m (= 0; < 0; > 0)$

is called as the magnetic susceptibility of the medium

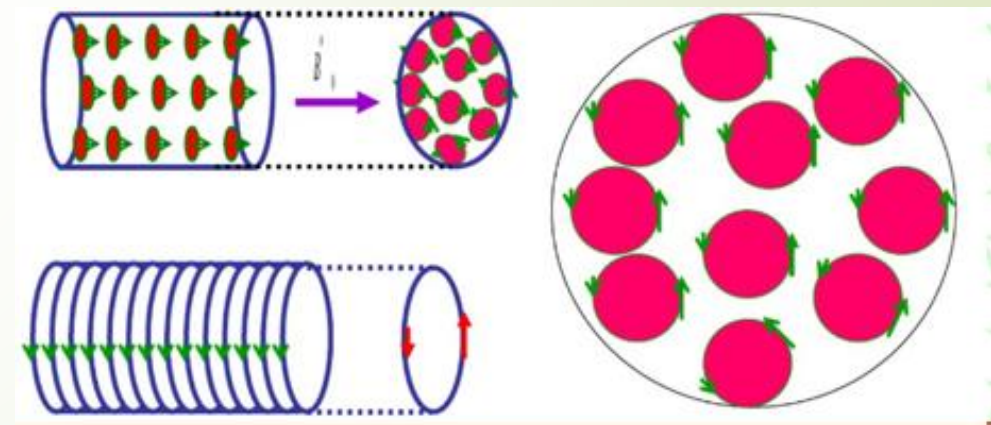
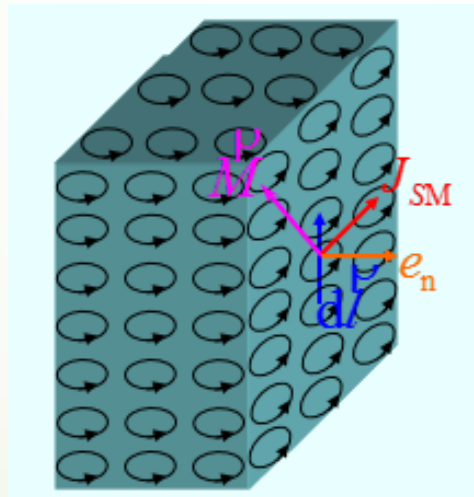
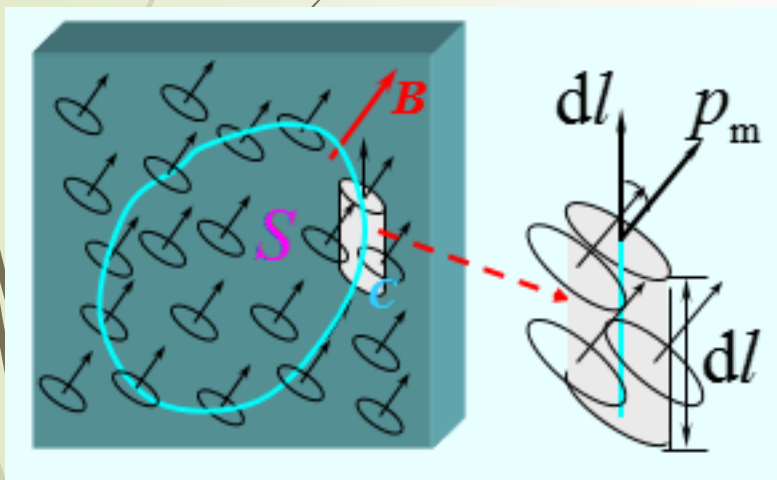
- Magnetic flux density on the axis of the current carrying circular ring :

$$\begin{aligned}\vec{B}(0,0,z) &= \vec{e}_z \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}} \\ &= \frac{\mu_0 \vec{p}_m}{2\pi(a^2 + z^2)^{3/2}}\end{aligned}$$



Further discussion on magnetization phenomenon

- After medium is magnetized, net current distribution $\vec{J}_M$ could appear **internally**, that is the generation of volume magnetization current.
- After medium is magnetized, net current distribution $\vec{J}_{SM}$ could appear **at interface of medium**, that is the generation of surface magnetization current.



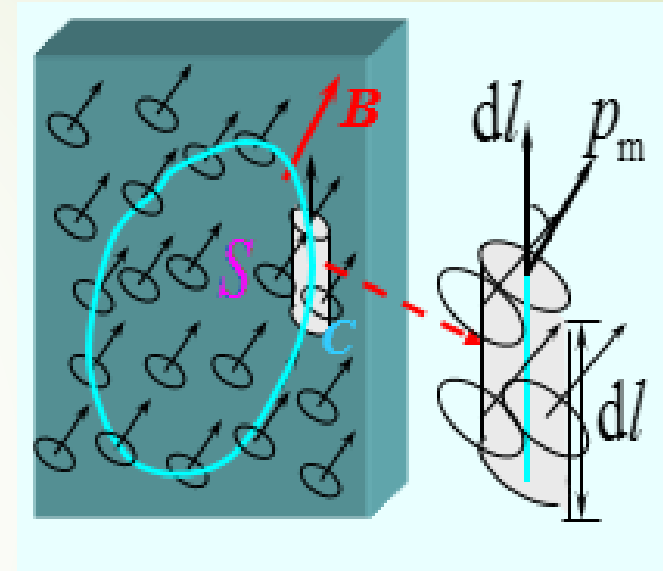
Calculation of volume magnetic current density

Calculation Principle : $I_M|_S = I_M|_C$

$$\begin{aligned}\text{As, } I_M|_{dC} &= iN_{dC} = indV = ni\Delta S dl \cos \theta \\ &= np_m dl \cos \theta = M dl \cos \theta = \vec{M} \cdot d\vec{l}\end{aligned}$$

so,
$$I_M = \oint_C \vec{M} \cdot d\vec{l}$$

→
$$\vec{J}_M = \nabla \times \vec{M}$$



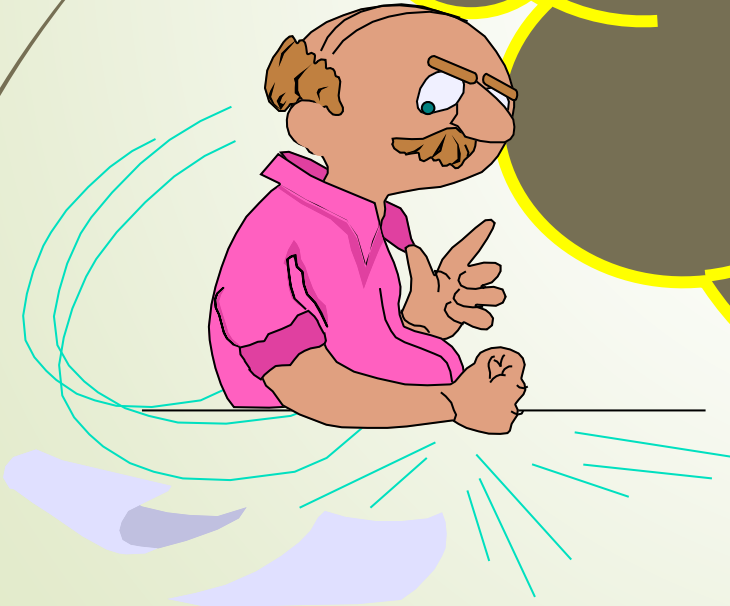
Calculation of surface magnetic current density

At the interface of media :

$$\text{As, } I_M|_{dC} = \vec{M} \cdot d\vec{l}$$

$$\text{so, } I_M|_{dC} = M_t dl \qquad J_{SM} = M_t$$

→
$$\vec{J}_{SM} = \vec{M} \times \vec{e}_n$$



Question: When magnetized current exist, **additional magnetic field will be generated** and affect the total magnetic field distribution. Then when we calculate the total field, is it necessary to calculate the magnetic field generated by the magnetic current **in advance**?

6.5 Magnetic Field Intensity and Relative Permeability

❖ The macroscopic effect of magnetization can be studied by incorporating the equivalent volume current density.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S (\vec{J} + \vec{J}_M) \cdot d\vec{S} \quad \longrightarrow \quad \begin{aligned} \nabla \times \vec{B} &= \mu_0 (\vec{J} + \vec{J}_M) \\ \nabla \times \vec{B} &= \mu_0 (\vec{J} + \nabla \times \vec{M}) \\ \nabla \times \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right) &= \vec{J} \end{aligned}$$

Introduce magnetic field intensity:

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \quad , \quad \text{so } \vec{B} = \mu_0 (\vec{H} + \vec{M})$$

then $\nabla \times \vec{H}(\vec{r}) = \vec{J}(\vec{r})$

The integral form is $\oint_C \vec{H}(\vec{r}) \cdot d\vec{l} = \int_S \vec{J}(\vec{r}) \cdot d\vec{S}$



Conclusion

When there are medium in space , the problems of **steady magnetic field** can be described by the following **basic equations**.

$$\begin{cases} \nabla \times \vec{H}(\vec{r}) = \vec{J}(\vec{r}) \\ \nabla \cdot \vec{B}(\vec{r}) = 0 \end{cases}$$

(Differential form)

$$\begin{cases} \oint_C \vec{H}(\vec{r}) \cdot d\vec{l} = \int_S \vec{J}(\vec{r}) \cdot d\vec{S} \\ \oint_S \vec{B}(\vec{r}) \cdot d\vec{S} = 0 \end{cases}$$

(integral form)

The process of solving the problems can use following route :

$$\vec{J} \longrightarrow \vec{H} \longrightarrow \vec{B}, \vec{M}, \vec{J}_M$$

Ex. There is an infinite conducting magnetic cylinder, with permeability of μ and radius of a , and a line current I exist on the axis of it. Outside the cylinder is air (μ_0), find the distribution of $\vec{B}$, $\vec{H}$ and $\vec{M}$ outside the cylinder.

Solution: Applying Ampere's law, then

$$\oint_C \vec{H} \cdot d\vec{l} = 2\pi\rho H_\phi = I$$

Magnetic
field intensity

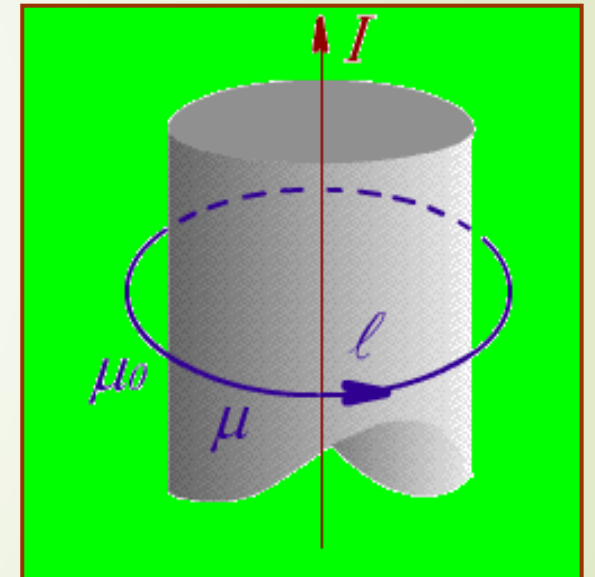
$$\vec{H} = \vec{e}_\phi \frac{I}{2\pi\rho} \quad 0 < \rho < \infty$$

Magnetic
flux density

$$\vec{B} = \begin{cases} \vec{e}_\phi \frac{\mu I}{2\pi\rho} & 0 < \rho < a \\ \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho} & a < \rho < \infty \end{cases}$$

Magnetization
intensity

$$\vec{M} = \frac{\vec{B}}{\mu_0} - \vec{H} = \begin{cases} \vec{e}_\phi \frac{\mu - \mu_0}{\mu_0} \cdot \frac{I}{2\pi\rho} & \rho < a \\ 0 & a < \rho < \infty \end{cases}$$



• Classification and constitutive relation of magnetic medium

Classification :

$$\mu_r < 1$$

Diamagnetic
substance

Water: 0.99999

$$\mu_r > 1$$

Paramagnetic
substance

Air: 1.0000004

$$\mu_r \gg 1$$

Ferromagnetic
substance

Iron: 4000

Constitutive relation :

$$\vec{M} = \chi_m \vec{H} \quad \Rightarrow \quad \vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$$

Magnetic
susceptibility
(dimensionless)

Magnetic permeability

$$\vec{B} = \mu \vec{H}$$

Relative magnetic
permeability
(dimensionless)



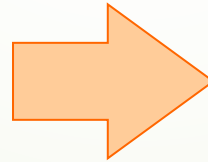
6.6 Boundary Condition for Magnetostatic Fields

1. Magnetic flux density B

Choose a point P on the interface, make a cylindrical surface S enclosed the point P

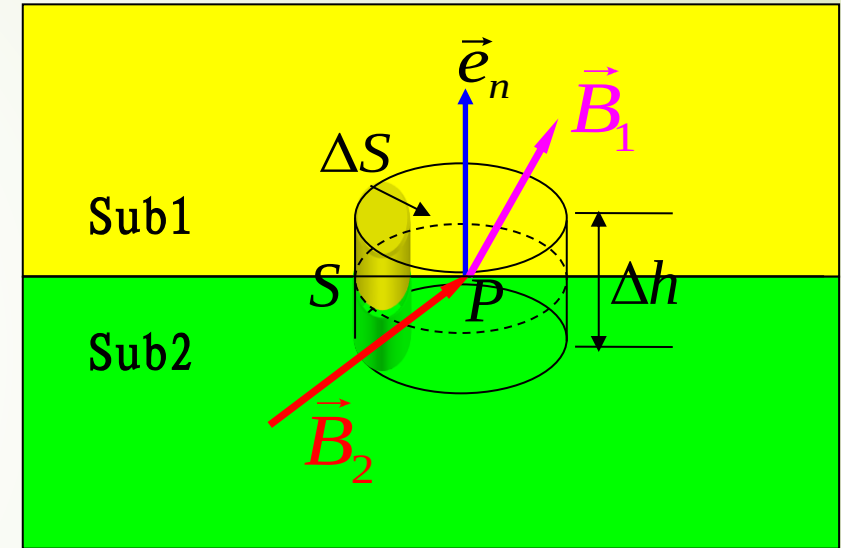
Since $\Delta h \rightarrow 0$, then

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$



$$(\vec{B}_1 - \vec{B}_2) \cdot \vec{e}_n \Delta S = 0$$

So $\vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$ Or $B_{1n} = B_{2n}$



2. Magnetic field intensity H

Choose the closed path as the contour C at the both sides of the interface, Since $\Delta h \rightarrow 0$, then

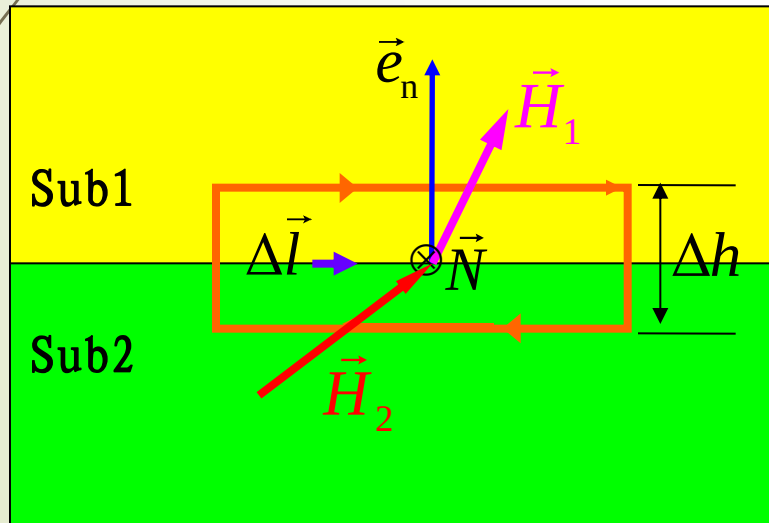
$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S} \quad \Rightarrow (\vec{H}_1 - \vec{H}_2) \cdot \Delta \vec{l} = \vec{J}_s \cdot \vec{N} \Delta l$$

$$\Delta \vec{l} = \vec{N} \times \vec{e}_n \Delta l \quad \Rightarrow (\vec{H}_1 - \vec{H}_2) \cdot \Delta \vec{l} = (\vec{H}_1 - \vec{H}_2) \cdot (\vec{N} \times \vec{e}_n) \Delta l$$

$$= [\vec{e}_n \times (\vec{H}_1 - \vec{H}_2)] \cdot \vec{N} \Delta l$$

So $\vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$

Or $H_{1t} - H_{2t} = J_s$



➤ Two perfect dielectric

$$\begin{cases} \vec{e}_n \cdot (\vec{D}_1 - \vec{D}_2) = 0 \\ \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \\ \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = 0 \end{cases}$$

❖ Perfect dielectric and perfect conductor

$$\begin{cases} \vec{e}_n \cdot \vec{D} = \rho_s \\ \vec{e}_n \cdot \vec{B} = 0 \\ \vec{e}_n \times \vec{E} = 0 \\ \vec{e}_n \times \vec{H} = \vec{J}_s \end{cases}$$



➤ Boundary conditions of Magnetic vector potential

$$\vec{A}_1 = \vec{A}_2$$

$$\vec{e}_n \times \left(\frac{1}{\mu_1} \nabla \times \vec{A}_1 - \frac{1}{\mu_2} \nabla \times \vec{A}_2 \right) = \vec{J}_s$$

$$\nabla^2 \vec{A} = -\mu \vec{J}$$

$$\oint_C \vec{A} \cdot d\vec{l} = \int_S \vec{B} \cdot d\vec{S}$$

$$\nabla \cdot \vec{A} = 0$$

$$\vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

$$\vec{H} = \nabla \times \vec{A} / \mu$$

$$\Rightarrow A_{1t} = A_{2t}$$

$$\Rightarrow A_{1n} = A_{2n}$$

$$\Rightarrow \vec{A}_1 = \vec{A}_2$$

$$\left. \begin{array}{l} \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \\ \vec{H} = \nabla \times \vec{A} / \mu \end{array} \right\} \Rightarrow \vec{e}_n \times \left(\frac{1}{\mu_1} \nabla \times \vec{A}_1 - \frac{1}{\mu_2} \nabla \times \vec{A}_2 \right) = \vec{J}_s$$

Ex. The air between two conductor plate ($z = 0$ and $z = d$), the electric field intensity is known as

$$\vec{E} = \vec{e}_y E_0 \sin\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) \quad \text{V/m}$$

Find: (1) Magnetic field intensity $\mathbf{H}$;

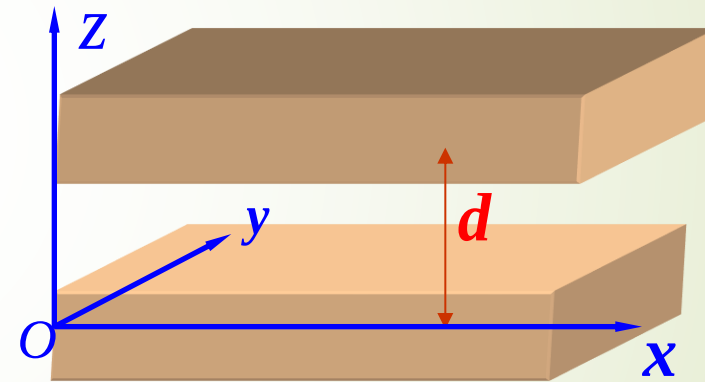
(2) The surface current density $\mathbf{J}_s$.

Solution (1) As $\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$

$$\frac{\partial \vec{H}}{\partial t} = -\frac{1}{\mu_0} \nabla \times \vec{E}$$

$$= -\frac{1}{\mu_0} \left(-\vec{e}_x \frac{\partial E_y}{\partial z} + \vec{e}_z \frac{\partial E_y}{\partial x} \right)$$

$$= \frac{E_0}{\mu_0} \left[\vec{e}_x \frac{\pi}{d} \cos\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) - \vec{e}_z k_x \sin\left(\frac{\pi}{d} z\right) \sin(\omega t - k_x x) \right]$$



Calculate the time integral of above equation

$$\begin{aligned}\vec{H}(x, z, t) &= \int \frac{\partial \vec{H}(x, z, t)}{\partial t} dt \\ &= \vec{e}_x \frac{\pi E_0}{\omega \mu_0 d} \cos\left(\frac{\pi}{d} z\right) \sin(\omega t - k_x x) + \\ &\quad \vec{e}_z \frac{k_x E_0}{\omega \mu_0} \sin\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) \quad (\text{A/m})\end{aligned}$$

(2) The current density of the conductor surface at $z = 0$

$$\vec{J}_S = \vec{e}_z \times \vec{H} \Big|_{z=0} = \vec{e}_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \quad (\text{A/m})$$

The current density of the conductor surface at $z = d$

$$\vec{J}_S = (-\vec{e}_z) \times \vec{H} \Big|_{z=d} = \vec{e}_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \quad (\text{A/m})$$

6.7 Inductances and Inductors

1. Magnetic Flux and Flux linkage

- Magnetic Flux (磁通)

$$\Phi = \int_S \vec{B} \cdot d\vec{S} = \int_S \nabla \times \vec{A} \cdot d\vec{S} = \oint_C \vec{A} \cdot d\vec{l}$$

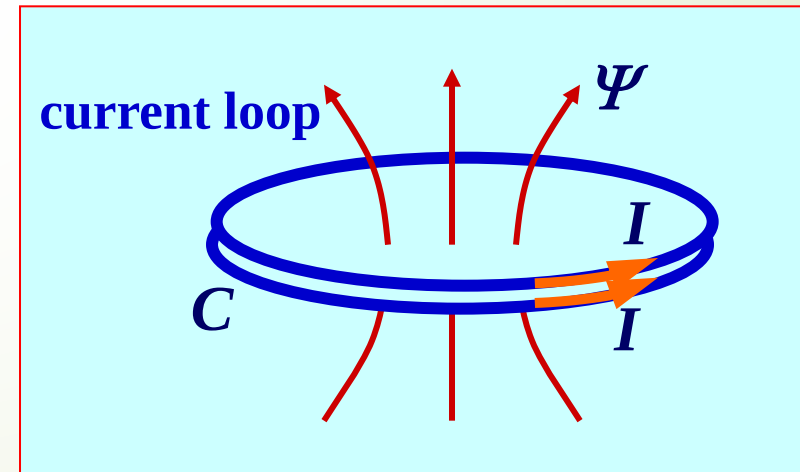
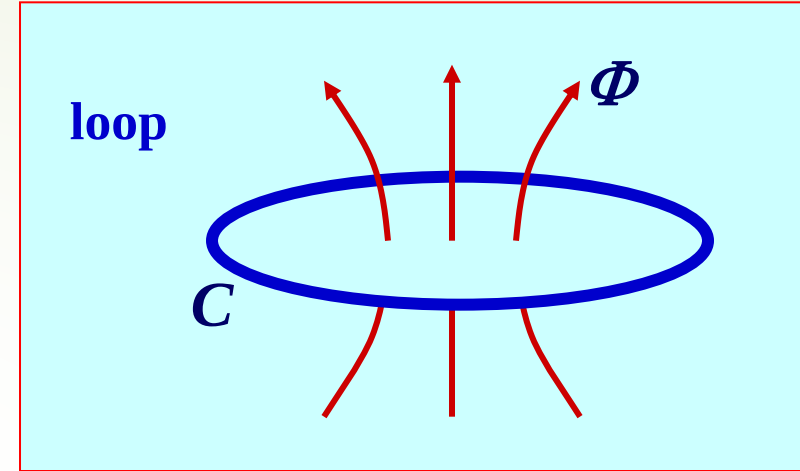
Features: the circuit can be any geometric loops

- Flux linkage (磁链)

Total flux of all current loop hinges

Features:

- The loop is current loop
- Include all the current loops
- Each loop includes the total magnetic flux linkage of the hinge



● Magnetic flux calculation

✓ Single turn coil

$$\Psi = \Phi$$

✓ Multi turn coil

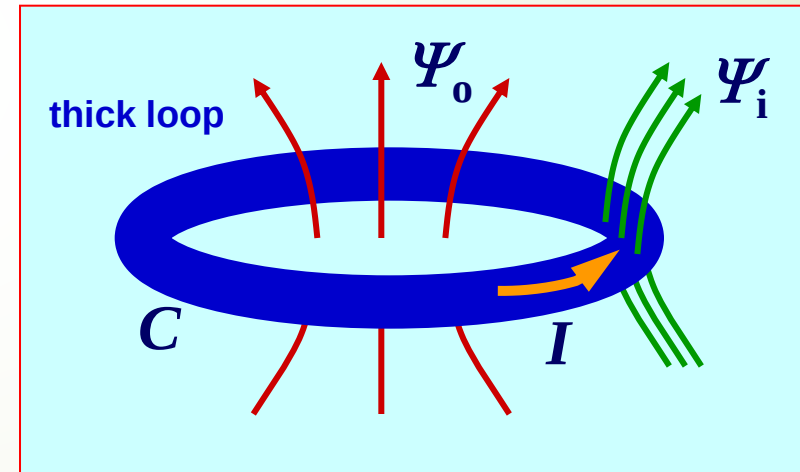
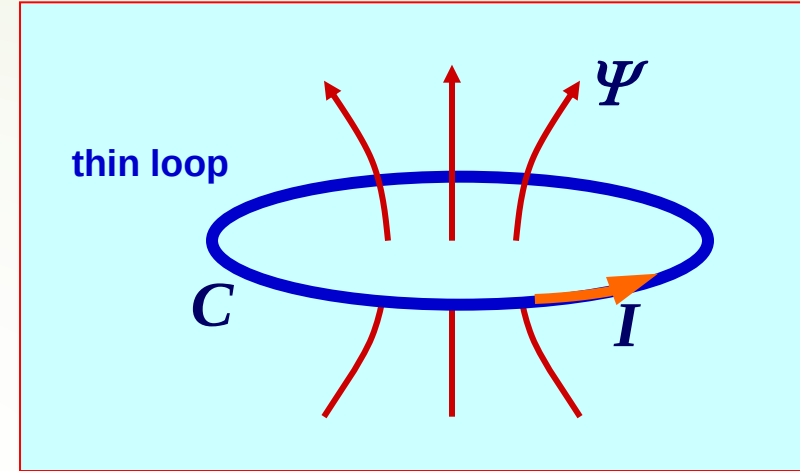
$$\Psi = n\Phi$$

n : is the ratio the current of the magnetic field hinge to the current of the circuit current I (Not necessarily integer)

✓ Thick wire loops

$$\Psi = \Psi_o + \Psi_i$$

Ψ_o : External magnetic flux ; Ψ_i :Internal magnetic flux



2. Self-inductance

$$L = \frac{\Psi}{I}$$

called the self inductance coefficient of circuit C, short for self-inductance .

Features: the magnetic flux is generated by I.

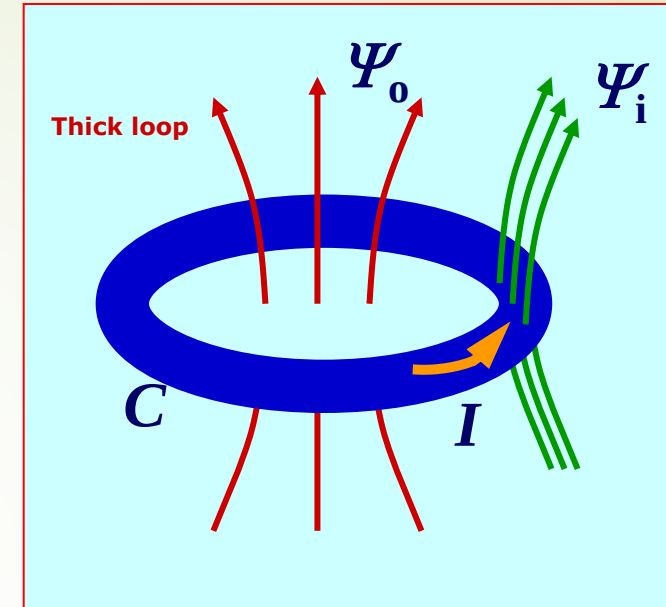
Self-inductance of thick conductor loop :

$$L = L_i + L_o$$

$$L_i = \frac{\Psi_i}{I} \quad \text{—— Internal self inductance ;} \quad L_o = \frac{\Psi_o}{I} \quad \text{—— External self inductance ;}$$

■ **The characteristics of self-inductance:**

self-inductance is related to geometry of loops , size and the surrounding magnetic medium, and not related the value of the magnetic flux and current .



3. Mutual inductance

$$M_{21} = \frac{\Psi_{21}}{I_1} \quad (M_{11} = L_1)$$

Mutual inductance coefficient of loop C_1 to loop C_2 , short for mutual inductance.

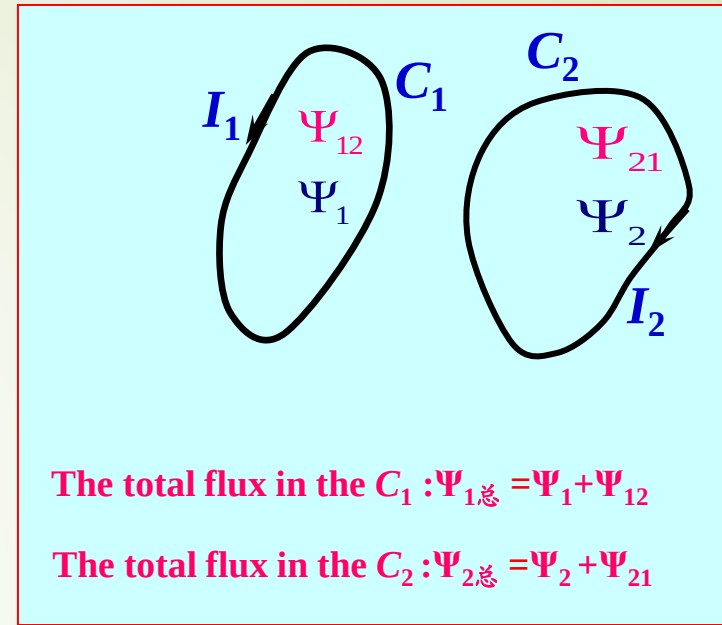
Features: the magnetic flux in the C_2 is generated by I_1

Similarly, the mutual inductance of the loop C_2 to the loop C_1 is

$$M_{12} = \frac{\Psi_{12}}{I_2} <$$

Features: the magnetic flux in the C_1 is generated by I_2

Question : $\Psi_{1\text{total}} = ?$; $\Psi_{2\text{total}} = ?$



$$M_{21} = M_{12} = M$$

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{dl_1 \cdot dl_2}{R}$$

Neumann's
formula

4. Proof of the Newman's formula

$$\Psi_{21} = \oint_{C_2} \vec{A}_1 \cdot d\vec{l}_2$$

And: $\vec{A}_1(\vec{r}_2) = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\vec{l}_1}{R}$

Then:

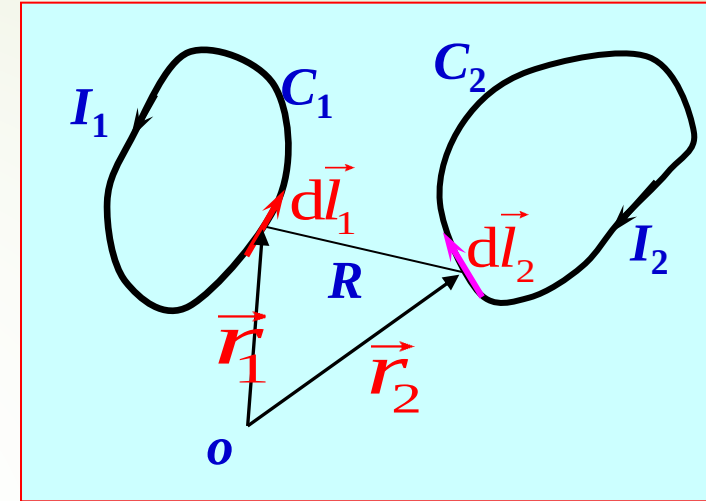
$$\Psi_{21} = \oint_{C_2} \vec{A}_1 \cdot d\vec{l}_2 = \frac{\mu_0 I_1}{4\pi} \int_{C_2} \oint_{C_1} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$

**Newman's
formula**

So: $M_{21} = \frac{\Psi_{21}}{I_1} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$

Similarly: $M_{12} = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\vec{l}_2 \cdot d\vec{l}_1}{R} = M_{21}$

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$



Ex. Find the self-inductance for coaxial line.

Solution: find the internal self-inductance first

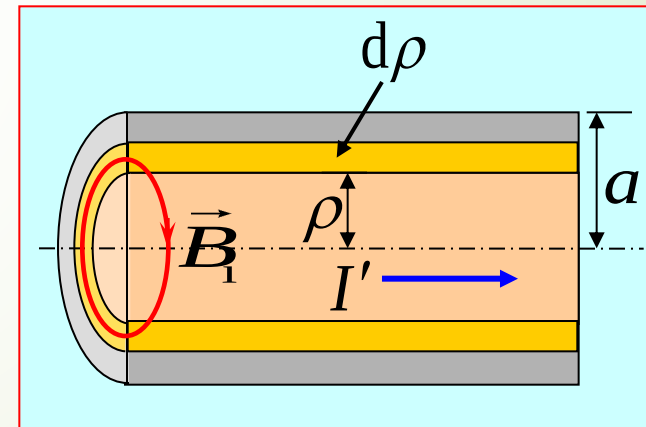
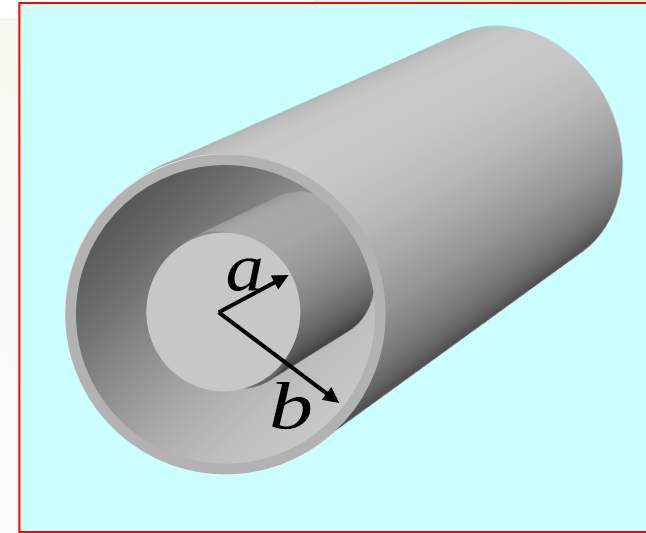
$$\oint_C \vec{H}_i \cdot d\vec{l} = I' = \frac{I}{\pi a^2} \pi \rho^2 = \frac{I}{a^2} \rho^2$$

so
$$\vec{H}_i = \frac{I}{2\pi a^2} \rho, \quad \vec{B}_i = \frac{\mu_0 I}{2\pi a^2} \rho$$

Magnetic Flux for surface element at ρ

$$d\Phi_i = \vec{B}_i \cdot d\vec{S} = \frac{\mu_0 I}{2\pi a^2} \rho l d\rho$$

Flux linkage:
$$d\Psi_i = \frac{I'}{I} d\Phi_i = \frac{\mu_0 I \rho^3}{2\pi a^4} l d\rho$$



Total flux linkage in inner conductor is

$$\Psi_i = \int d\Psi_i = l \int_0^a \frac{\mu_0 I \rho^3}{2\pi a^4} d\rho = \frac{\mu_0 I}{8\pi} l$$

**internal self-inductance
per unit length**

$$L_i = \frac{\Psi_i}{Il} = \frac{\mu_0}{8\pi}$$

Find the external self-inductance between two conductors

$$B = \frac{\mu_0 I}{2\pi\rho} \quad \longrightarrow \quad d\Psi_o = d\Phi_o = \frac{\mu_0 I}{2\pi\rho} l d\rho$$

so

$$\Psi_o = \int d\Psi_o = l \int_a^b \frac{\mu_0 I}{2\pi\rho} d\rho = l \frac{\mu_0 I}{2\pi} \ln \frac{b}{a}$$

**External self-inductance
per unit length**

$$L_o = \frac{\Psi_o}{Il} = \frac{\mu_0}{2\pi} \ln \frac{b}{a}$$

**Total self-inductance
per unit length**

$$L = L_i + L_o = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a}$$



Ex. Find the self-inductance per unit length of a two-wire transmission line. ($D \gg a$)

Solution: Applying Ampere's circuital law and superposition

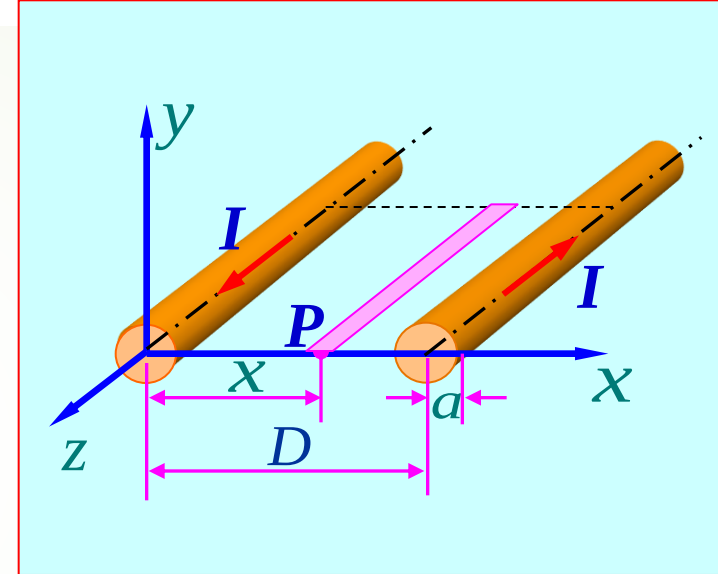
$$\vec{B}(x) = \vec{e}_y \frac{\mu_0 I}{2\pi} \left(\frac{1}{x} + \frac{1}{D-x} \right)$$

External flux linkage,

$$\Psi_o = \int \vec{B} \cdot d\vec{S} = \frac{\mu_0 I}{2\pi} l \int_a^{D-a} \left(\frac{1}{x} + \frac{1}{D-x} \right) dx = \frac{\mu_0 I}{\pi} l \ln \frac{D-a}{a}$$

External self-inductance per unit length:

$$L_o = \frac{\Psi_o}{I} = \frac{\mu_0}{\pi} \ln \frac{D-a}{a} \approx \frac{\mu_0}{\pi} \ln \frac{D}{a}$$



internal self-inductance per unit length for two wires

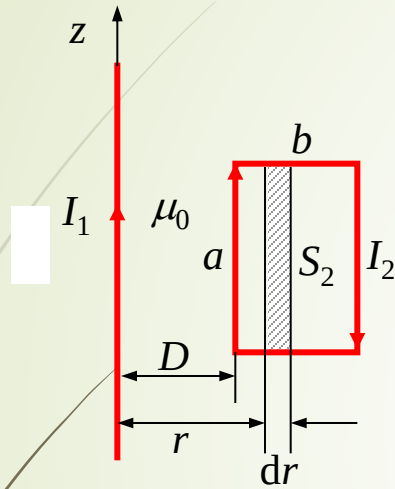
$$L_i = 2 \times \frac{\mu_0}{8\pi} = \frac{\mu_0}{4\pi}$$

The total self-inductance per unit length of a two-wire transmission line

$$L = L_i + L_o = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln \frac{D}{a}$$



Ex. Calculate the mutual inductance between an infinitely long straight line and a rectangular coil. The line and the coil are at the same plane, and in vacuum.



Solution: Select cylindrical coordinate system, and let the infinitely long straight line to be at the z -axis. The magnetic flux density produced by current I_1 is then

$$\mathbf{B}_1 = \frac{\mu_0 I_1}{2\pi r} \mathbf{e}_\phi$$

The magnetic flux linkage Ψ_{21} with current I_2 by current I_1 is

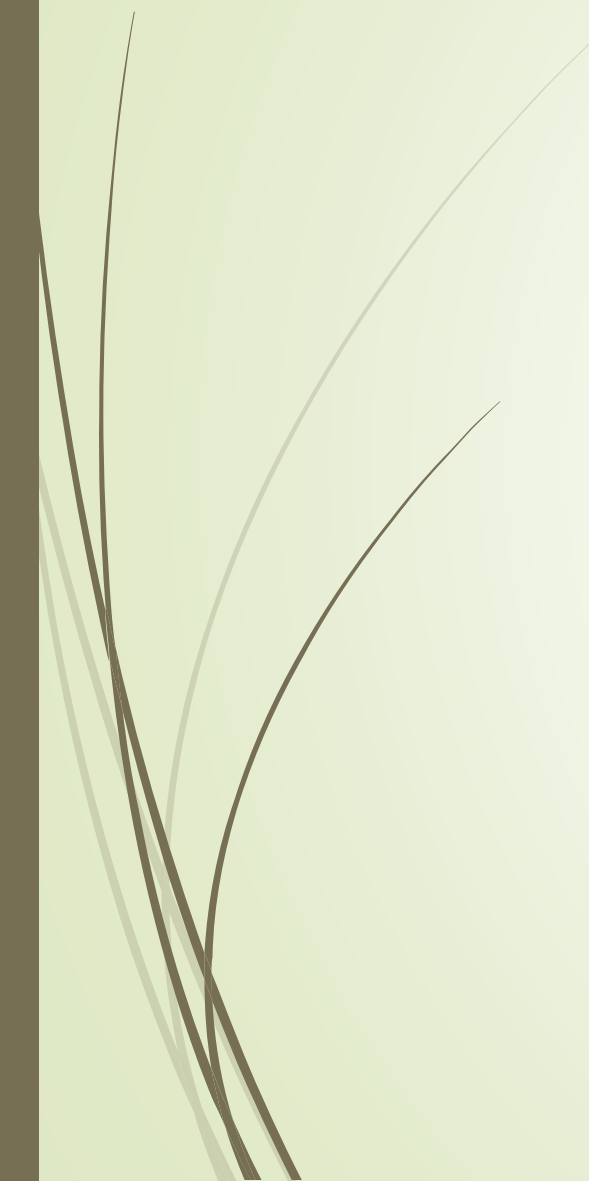
$$\Psi_{21} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{S}$$

If the flowing direction of the current I_2 is clockwise, $d\mathbf{S}$ and $\mathbf{B}_1$ have the same direction. Then

$$\Psi_{21} = \frac{\mu_0 I_1 a}{2\pi} \int_D^{D+b} \frac{1}{r} dr = \frac{\mu_0 I_1 a}{2\pi} \ln\left(\frac{D+b}{D}\right)$$

We have
$$M_{21} = \frac{\Psi_{21}}{I_1} = \frac{\mu_0 a}{2\pi} \ln\left(\frac{D+b}{D}\right) > 0$$

If the flowing direction of the current I_2 is **counter** clockwise, then the $\mathbf{B}_1$ and $d\mathbf{S}$ are opposite, and $M_{21} < 0$.



https://www.bilibili.com/video/BV1t64y1C7o7/?spm_id_from=333.337.search-card.all.click&vd_source=cbc26ac5a27dd541735237c7e8b5a6a4

6.8 Magnetic Energy

- Experimental evidence of magnetic field energy

$$\vec{F} = \oint_C \vec{I} d\vec{l} \times \vec{B}$$

- Distribution space of magnetic field energy

Where there is a magnetic field, there is the energy of the magnetic field !



1. Calculation of magnetic field energy by magnetic field distribution

- Energy density of magnetic field :

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H}$$

- Energy of magnetic field :

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV$$

The magnetic field energy
in the integral domain V

- For linear, isotropic medium it can be written as:

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{1}{2} \mu \vec{H} \cdot \vec{H} = \frac{1}{2} \mu H^2$$

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV = \frac{1}{2} \int_V \mu \vec{H} \cdot \vec{H} dV = \frac{1}{2} \int_V \mu H^2 dV$$



2. Calculation of magnetic field energy by magnetic vector potential

✓ Volume current distribution

$$W_m = \frac{1}{2} \int_V \vec{J} \cdot \vec{A} dV$$

✓ Surface current distribution

$$W_m = \frac{1}{2} \int_S \vec{J}_s \cdot \vec{A} dS$$

✓ Line current of a circuit loop

$$W_m = \frac{I}{2} \oint_C \vec{A} \cdot d\vec{l} = \frac{I}{2} \psi_{\text{total}}$$



3. Calculation of magnetic field energy by inductance

- Inductive energy storage

(Single current-carrying loop) :

$$W_m = \frac{I}{2} \psi_{\text{总}} = \frac{1}{2} I^2 L$$

- Inductive energy storage

(N current-carrying loops) :

$$\begin{aligned} W_m &= \frac{1}{2} \sum_{j=1}^N I_j \Psi_{j\text{总}} = \frac{1}{2} \sum_{j=1}^N I_j \sum_k^N \Psi_{jk} \\ &= \frac{1}{2} \sum_{j=1}^N \sum_k^N I_j I_k M_{jk} \end{aligned}$$

For example, two loops current-carrying

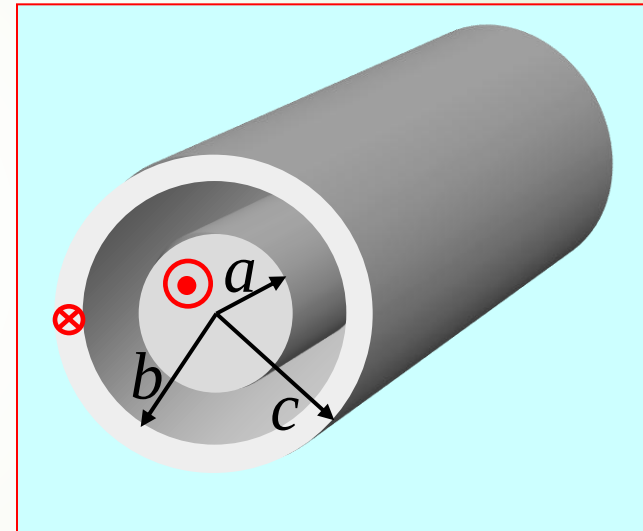
$$W_m = \frac{1}{2} \sum_{j=1}^N \sum_k^N I_j I_k M_{jk} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$$



Ex. Find the magnetic energy and self inductance in a coaxial line per unit length.

Solution: from Ampere's circuital law

$$\vec{H} = \begin{cases} \vec{e}_\phi \frac{\rho I}{2\pi a^2} & 0 < \rho < a \\ \vec{e}_\phi \frac{I}{2\pi\rho} & a < \rho < b \\ \vec{e}_\phi \frac{I}{2\pi\rho} \frac{c^2 - \rho^2}{c^2 - b^2} & b < \rho < c \\ 0 & \rho > c \end{cases}$$



The magnetic energy per unit length in three area:

$$W_{m1} = \frac{\mu_0}{2} \int_0^a \left(\frac{\rho I}{2\pi a^2} \right)^2 2\pi \rho d\rho = \frac{\mu_0 I^2}{16\pi}$$

$$W_{m2} = \frac{\mu_0}{2} \int_a^b \left(\frac{I}{2\pi \rho} \right)^2 2\pi \rho d\rho = \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a}$$

$$\begin{aligned} W_{m3} &= \frac{\mu_0}{2} \int_b^c \left(\frac{I}{2\pi \rho} \right)^2 \left(\frac{c^2 - \rho^2}{c^2 - b^2} \right)^2 2\pi \rho d\rho \\ &= \frac{\mu_0 I^2}{4\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right] \end{aligned}$$

The total magnetic energy per unit length

$$\begin{aligned} W_m &= W_{m1} + W_{m2} + W_{m3} \\ &= \frac{\mu_0 I^2}{16\pi} + \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a} + \frac{\mu_0 I^2}{4\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right] \end{aligned}$$

The total self-inductance per unit length

$$L = \frac{2W_m}{I^2} = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a} + \frac{\mu_0}{2\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right]$$

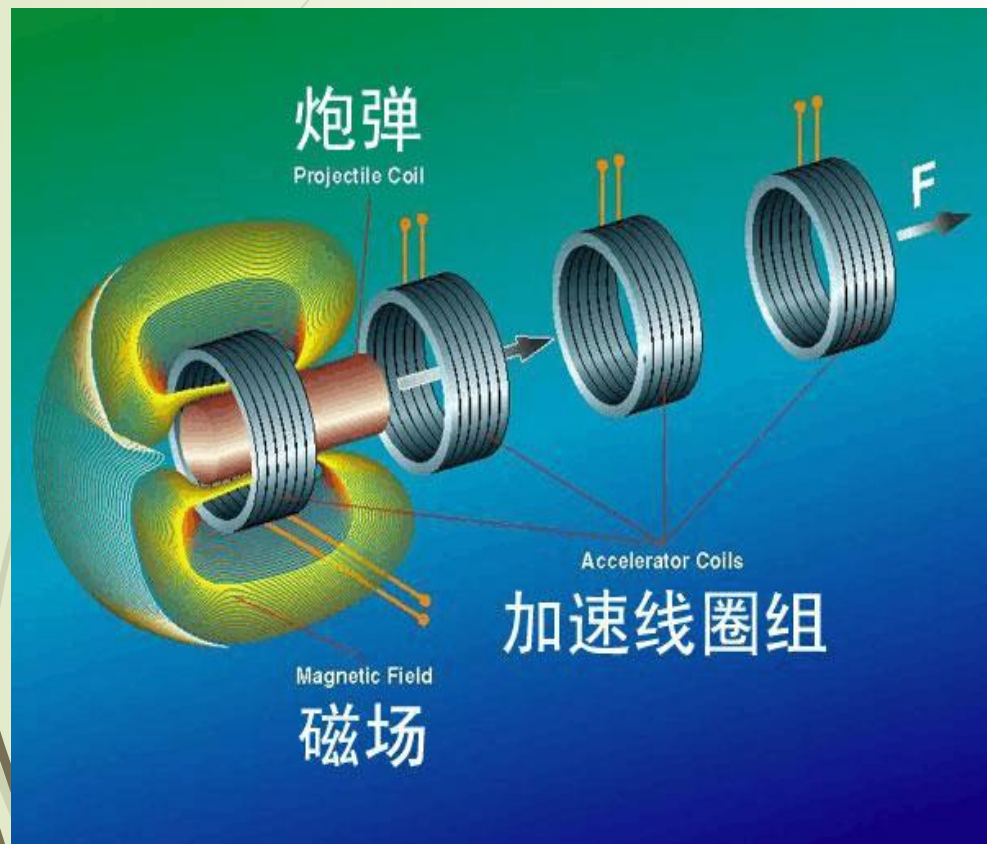
Internal self-inductance
for inner conductor

External self-inductance
between two conductors

Internal self-
inductance
for outer
conductor

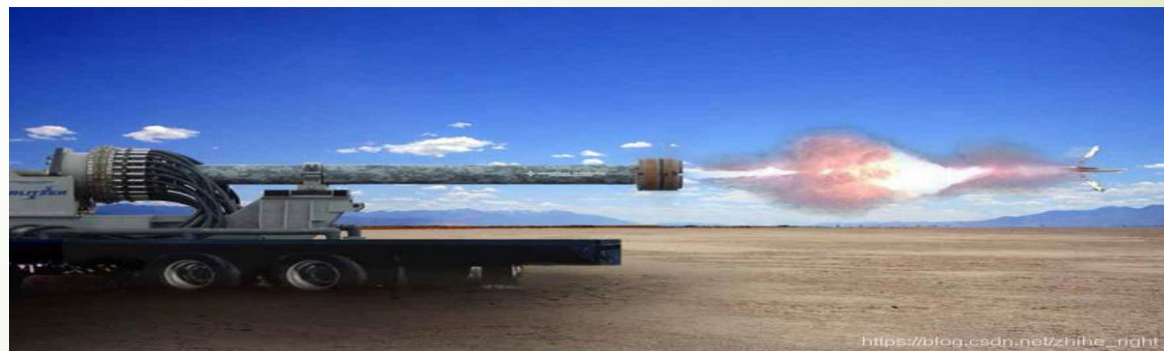
知识拓展：电磁炮

什么是电磁炮？



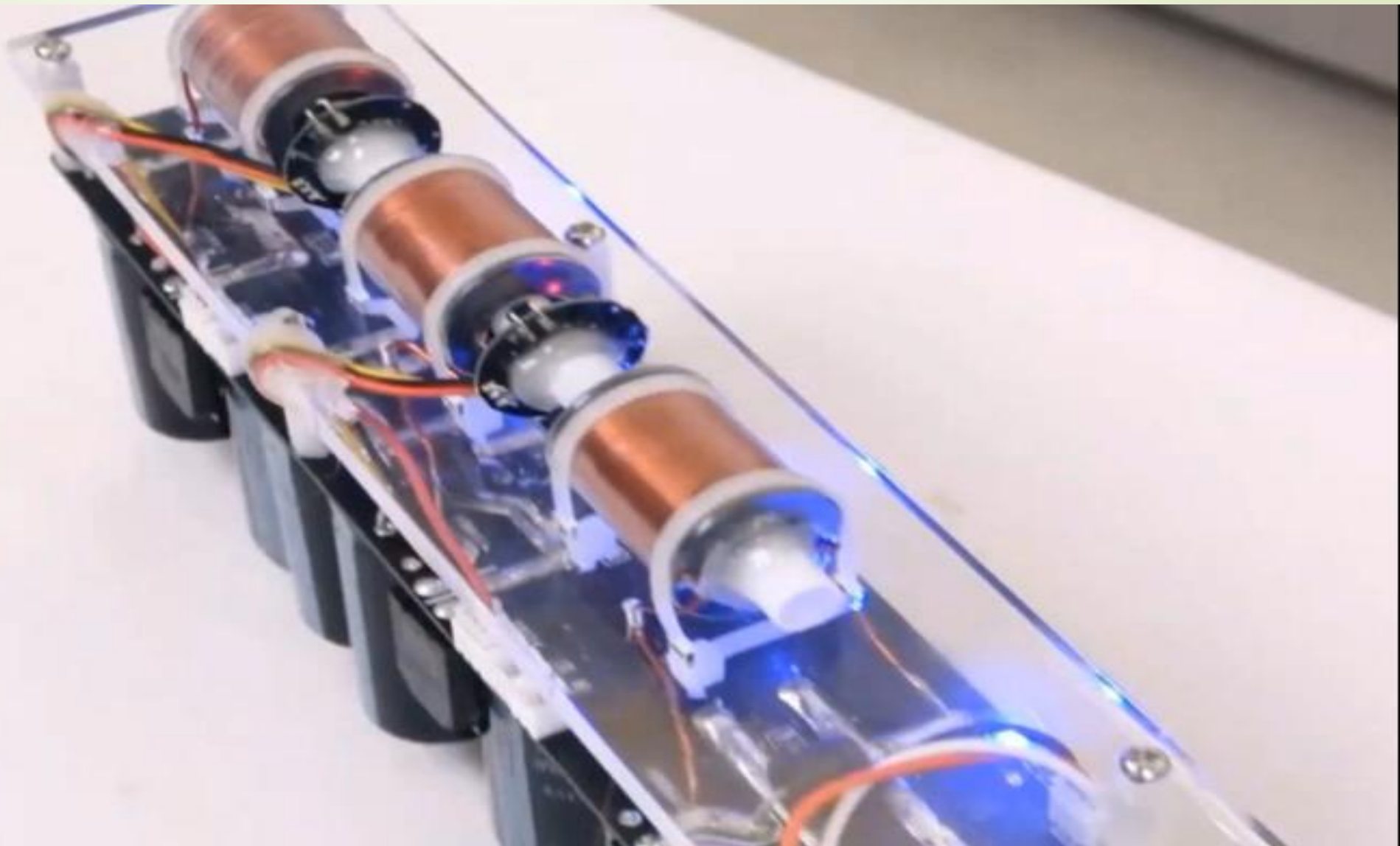
电磁炮有什么优点？

1. 弹丸初速高，射程远；
2. 结构简单，没有弹壳、药筒
3. 安全可靠，无炸药不会出现储存不当的爆炸。
4. 成本低：仅需电力。

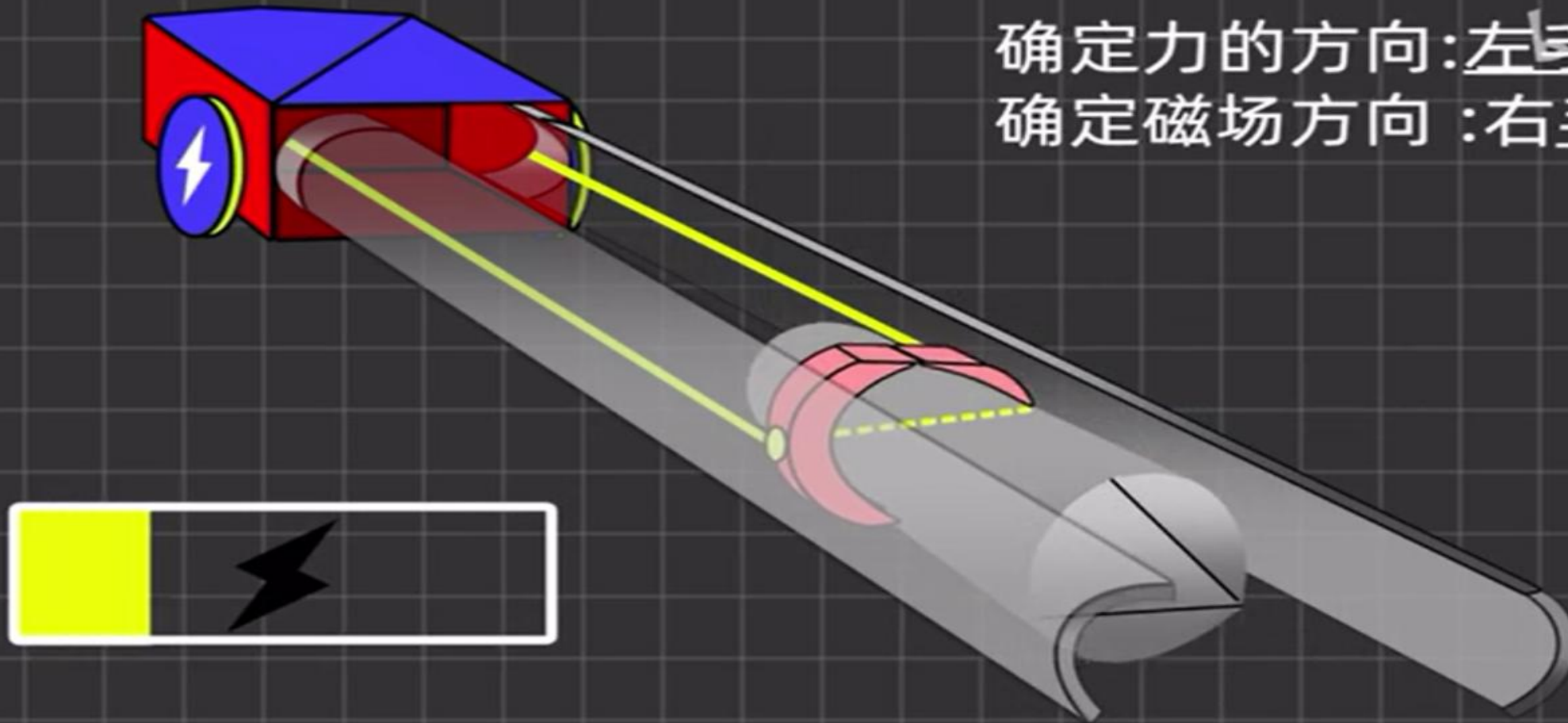


静电场的应用：电磁线圈炮（线圈间相互作用力）

@别小齐



静电场的应用：电磁轨道炮（安培力）



确定力的方向：左手定则
确定磁场方向：右手定则

只要通过的电流足够大

知识拓展：航母电磁弹射器

国别	舰种	舰级	现役数（艘）	方式	排水量（万吨）	可载战斗机数	航母战力评估
美国	重型核航母	福特级	1	电磁	10	60+	★★★★★+
	重型核航母	尼米兹级	10	蒸汽	9.5	60+	★★★★★
中国	重型航母	福建号	1	电磁	8+	40+	★★★★★
	中型航母	山东号	1	滑越	6	24+	★★★
	中型航母	辽宁号	1	滑越	6	24+	★★★
英国	中型航母	伊丽莎白女王级	2	滑越	6.5	40	★★★★
法国	中型核航母	戴高乐号	1	蒸汽	4.2	36	★★★+
俄罗斯	中型航母	库兹涅佐夫号	1	滑越	6	24	★
印度	中型航母	维克拉马迪亚号	1	滑越	4.5	24	★
	中型航母	维特兰特号	1	滑越	4	24	★
意大利	轻型航母	加富尔号	1	滑越	2.7	16	★
	轻型航母	加里波第号	1	滑越	1.3	12	★
合计			22				

知识拓展：航母电磁弹射器

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滑越式：山东号航空母舰

优点：简单

缺点：倾斜甲板不能放飞机
可载飞机数少



尼米兹级航空母舰

英国人搞出了蒸汽弹射式。

结构复杂，英国人弃用，美国人发扬光大。

知识拓展：航母电磁弹射器

采用蒸汽弹射的国家

法国（夏尔戴高乐号，全套美国设备）

美国（10艘航母）

优点：

相对于滑越式能多装载飞机

起飞距离短

缺点：

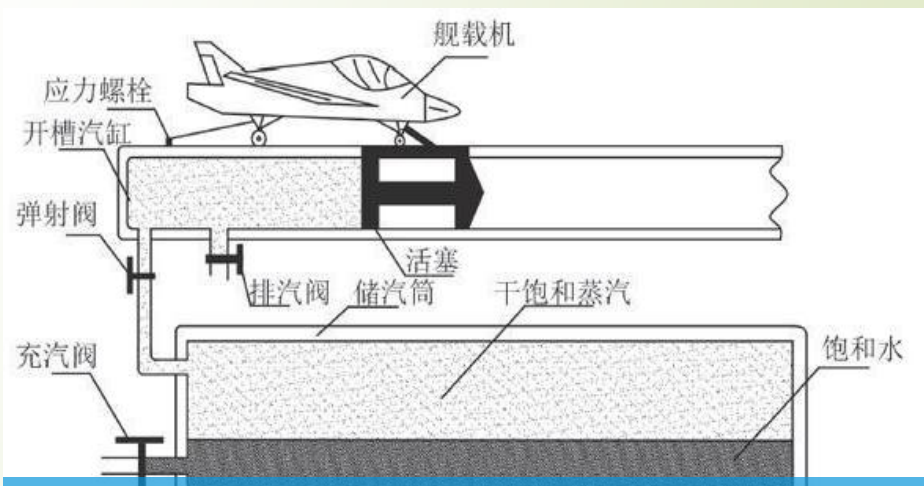
过载6G加速度

加速不均匀

无法调整弹射力度，不能适应多种机型



蒸汽管



蒸汽弹射原理

知识拓展：航母电磁弹射器

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采用电磁弹射的国家：

中国（福建号）

美国（福特级）

美国缺点：

使用交流电，不好调功率，线圈重，故障率高，发热量大，散热难。储存释放和全舰电力系统直接连通，易造成瞬间系统过载造成全舰断电。特朗普甚至扬言拆掉它，改装老版的蒸汽弹射器



福建号



福特级

知识拓展：航母电磁弹射器原理



原理与电磁轨道炮类似，实现的技术难度非常大。

电磁线圈炮弹射出去的是几十克的子弹或者几十公斤的炮弹。

电磁弹射器弹射出去的是几十吨的飞机。

知识拓展：航母电磁弹射器原理

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小红书

神都右威卫

小红书号:1091024212



Summary

1. Basic equations (general problems):

Differential forms :

$$\begin{cases} \nabla \times \vec{H} = \vec{J} \\ \nabla \cdot \vec{B} = 0 \end{cases}$$

Constitutive relations: $\vec{B} = \mu \vec{H}$

Integral forms:

$$\begin{cases} \oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S} \\ \oint_S \vec{B} \cdot d\vec{S} = 0 \end{cases}$$

2. Boundary conditions (general problems):

$$\begin{cases} \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \end{cases} \quad \text{OR} \quad \begin{cases} B_{1n} - B_{2n} = 0 \\ H_{1t} - H_{2t} = J_s \end{cases}$$

3. Magnetized current density:

$$\vec{J}_M = \nabla \times \vec{M} \quad \vec{J}_{SM} = \vec{M} \times \vec{e}_n$$

4. Inductance:

$$L = \frac{\Psi}{I} \quad L = L_i + L_o$$

5. Magnetic energy density:

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H}$$

6. Magnetic energy:

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV \quad W_m = \frac{1}{2} \int_V \vec{J} \cdot \vec{A} dV \quad W_m = \frac{I}{2} \psi_{\text{总}} = \frac{1}{2} I^2 L$$



Homework

According to the following link, explain the mechanism of Maglev train

https://www.bilibili.com/video/BV1sP4y1X7gX/?spm_id_from=333.337.search-card.all.click&vd_source=cbc26ac5a27dd541735237c7e8b5a6a4

 **6-4, 6-6, 6-10, 6-15, 6-22, 6-27, 6-36, 6-41**